

Relatório Bayes

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April 19, 2025

1 Analysis of Covariance Matrix Methods

This section evaluates three approaches for computing covariance matrices in a Gaussian Naive Bayes classifier: full covariance, naive (diagonal covariance), and complete-set covariance. Their effects on classification accuracy and likelihood space are discussed.

1.1 Methodologies

- **Full Covariance:** Computes class-specific covariance matrices using all feature interactions. Captures pairwise correlations within each class.
- **Naive Covariance:** Assumes feature independence by using diagonal covariance matrices (variances only). Ignores inter-feature correlations.
- **Complete-Set Covariance:** Derives a single covariance matrix from the entire dataset (ignoring class labels) and applies it to both classes. Reflects global correlations rather than class-specific ones.

1.2 Accuracy Results

The mean cross-validation accuracies (10-fold) for each method are:

- Reference (scikit-learn GaussianNB): 0.93 ± 0.03
- Full covariance: 0.94 ± 0.03
- Naive covariance: 0.90 ± 0.03
- Complete-set covariance: 0.92 ± 0.04

The full covariance model achieved the highest accuracy, likely due to its ability to model class-specific feature correlations. The naive approach underperformed, as expected from its independence assumption. The complete-set method showed intermediate performance, suggesting that global correlations partially approximate class-specific structures.

1.3 Likelihood Space Behavior

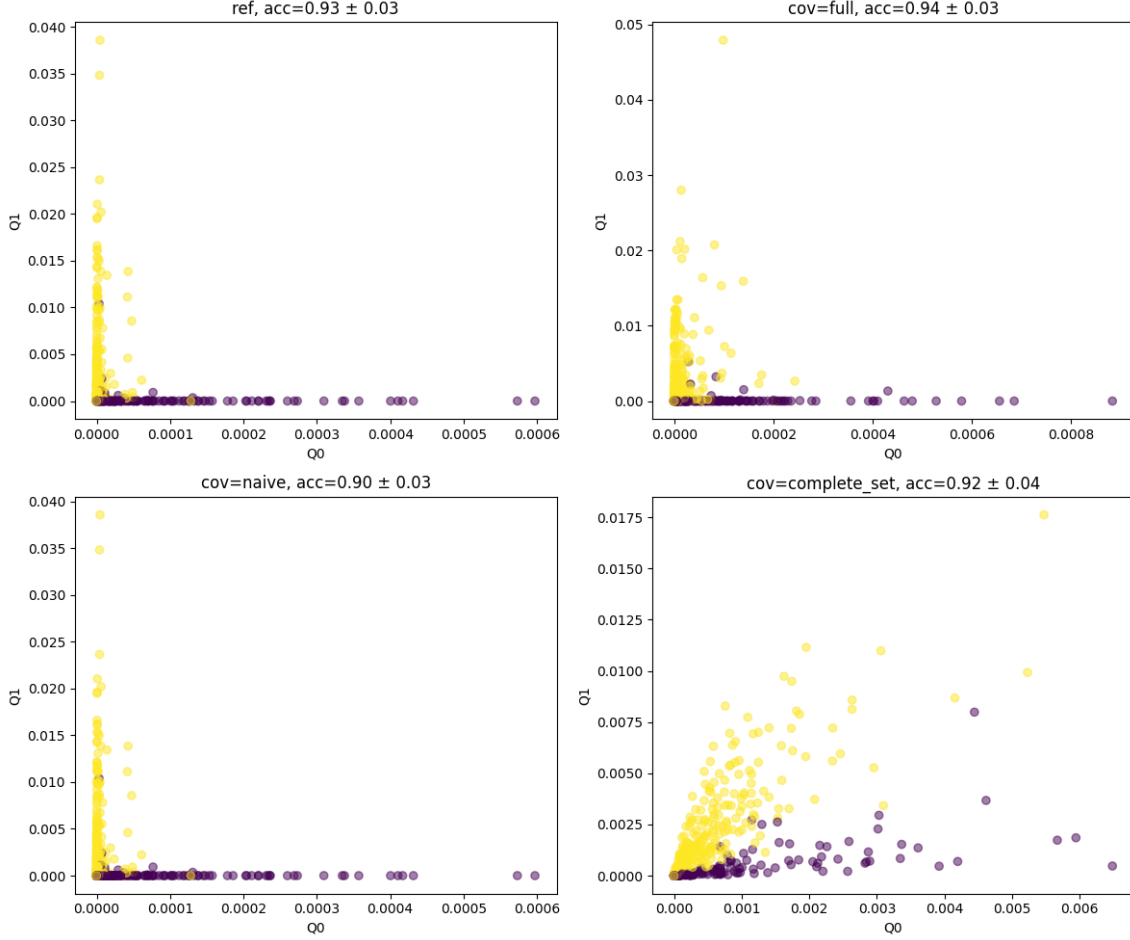


Figure 1: Likelihood spaces for different covariance methods. Points are colored by true class (0 or 1). The full covariance model (top-right) most closely resembles the reference model (top-left), with distinct class separation. The naive model (bottom-left) shows similar separation but lower accuracy, possibly due to unregularized variance estimates. The complete-set model (bottom-right) exhibits overlapping distributions, indicating mismatched covariance assumptions.

1.4 Discussion

The full covariance method outperformed others by leveraging class-specific correlations, aligning with the reference model’s behavior. However, its computational complexity increases with feature dimensionality. The naive approach suffered from oversimplification but remained robust to correlated features. The complete-set method’s intermediate accuracy suggests that global covariance partially compensates for class-specific correlations, but its divergent likelihood space implies suboptimal class separation. This contradiction warrants further investigation into dataset-specific correlation structures.